42X Academy Pvt. Ltd.
Office No. 15, 2nd Floor, Bhosale Shinde Arcade,
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Data Engineering — Full Stack Program

Industry-aligned journey from foundations to production-style pipelines

DURATION	PROGRAM TRACK
~19 weeks	Data Engineering

COURSE OVERVIEW

learners seeking a structured path into data engineering with Python, SQL, big data, cloud

WHAT YOU WILL LEARN

• Python and advanced SQL foundations.

• Warehousing, dimensional modeling and ETL/ELT.

• Apache Spark/PySpark for scale.

• Airflow orchestration and Azure Data Factory.

• dbt transformations and Kafka streaming fundamentals.

• Data quality, governance and security.

• End-to-end production-style capstone.

MODULE-WISE CURRICULUM

MODULE	TOPICS / LEARNING FOCUS	DURATION
1	Python for Data Engineering — scripting, data structures, APIs, files	2 weeks
2	SQL & Relational Databases — joins, CTEs, windows, optimization	2 weeks
3	Data Warehousing — dimensional modeling, star/snowflake, warehouse/lake/lakehouse	1.5 weeks
4	ETL/ELT Pipeline Design — extraction, transformation, loading, idempotency	1.5 weeks
5	Apache Spark (PySpark) — RDDs, DataFrames, Spark SQL, execution plans	2.5 weeks
6	Apache Airflow — DAGs, scheduling, dependencies, retries	1.5 weeks
7	Cloud Data Engineering — Azure Data Factory + AWS Glue/GCP BigQuery overview	2.5 weeks
8	dbt — models, tests, documentation	1 week
9	Kafka — producers, consumers, topics, real-time basics	1.5 weeks
10	Data Quality, Governance & Security — validation, lineage, access control, privacy	1 week

PRACTICAL SKILLS

Python scripting • Advanced SQL • Data modeling • ETL/ELT design • PySpark • Airflow • Azure Data Factory • dbt • Kafka • Data quality/governance

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PROJECT / CAPSTONE

Production-style end-to-end pipeline: ingestion → transformation → warehouse → dashboard-ready output, run through two-week Jira sprints with midpoint stakeholder review and final demo.

LEARNING OUTCOMES

- Build and explain end-to-end pipelines.
- Use Python and SQL as core tools.
- Process data at scale with Spark.
- Orchestrate workflows with Airflow.
- Build cloud workflows with ADF.
- Apply transformation, quality and governance practices.
- Deliver an industry-style capstone.

POTENTIAL CAREER PATHS

Data Engineer • Junior Data Engineer • Cloud Data Engineering roles • ETL / Data Pipeline roles

THE 42X DIFFERENCE

Combines SQL/Python foundations with Spark, one cloud platform, orchestration and transformation tooling. Azure is the default primary cloud in the source; AWS Glue and GCP BigQuery are introduced as alternatives.